

Important clarification: Despite the filename containing 'segmentation', this is a slice-level classification model, not a pixel-wise segmentation model. It outputs a single class label per CT slice (Background / Liver / HCC Tumor). Write accordingly in your paper.

1. Model Architecture

The proposed model uses MobileNet-V2 as a pretrained feature extractor (ImageNet weights), with its original classification head replaced by a custom two-layer fully connected network for 3-class HCC diagnosis from CT slices. The backbone consists of a Conv2d stem followed by 19 inverted residual bottleneck blocks using depthwise separable convolutions with ReLU6 activations, ending in a global average pooling layer producing a 1280-dimensional feature vector.

Input / Output Specification

Input shape	224 x 224 x 3 (grayscale CT slice stacked into 3 identical channels)
Output	3-class softmax logits: Background (0), Liver (1), HCC Tumor (2)
No skip connections	This is NOT a U-Net — no decoder, no upsampling, no pixel-level mask

Backbone — MobileNet-V2 (frozen structure, all layers trainable)

Layer	Kernel	Activation	Notes
Conv2d stem	3x3, stride 2	BN + ReLU6	32 filters
Inverted residual blocks x 19	1x1 + DW 3x3 + 1x1	BN + ReLU6 / Linear	Expanding then projecting
Global average pooling	—	—	1280-d feature vector

Custom Classifier Head (replaces original MobileNet-V2 head)

Layer	Shape	Activation	Notes
Dropout	p = 0.2	—	Regularization before projection
Linear	1280 → 256	—	Dimensionality reduction
ReLU	—	ReLU	Non-linearity
Dropout	p = 0.3	—	Stronger regularization before output
Linear	256 → 3	Softmax	Final class probabilities

Paper sentence: "The proposed model employs MobileNet-V2 as a pretrained feature extractor, with its original classification head replaced by a custom two-layer fully connected network incorporating dropout regularization for 3-class HCC diagnosis."

2. Hyperparameters

Hyperparameter	Value
Learning rate	0.001
Batch size	32
Epochs	15
Optimizer	Adam (torch.optim.Adam)
Loss function	CrossEntropyLoss
LR scheduler	ReduceLROnPlateau factor = 0.5 patience = 3 epochs
DataLoader workers	2

Paper sentence: "The model was trained using the Adam optimizer with a learning rate of 0.001 and a batch size of 32 over 15 epochs. A ReduceLROnPlateau scheduler halved the learning rate when validation loss plateaued for 3 consecutive epochs."

3. Data Preprocessing

The following pipeline is applied to every CT slice before model input. No preprocessing is skipped at validation or test time (except augmentation).

Step	Description
Step 1 — HU Windowing	Clip raw Hounsfield Units to [-75, 150] then normalize to [0, 1]. This liver-specific window makes soft tissue and HCC tumors maximally distinguishable.
Step 2 — CLAHE	Contrast Limited Adaptive Histogram Equalization (clip limit = 0.01). Enhances local contrast and tumor boundary visibility without amplifying noise.
Step 3 — Resize	Resize to 224 x 224 pixels to match MobileNet-V2 input requirements.
Step 4 — RGB conversion	Stack the grayscale channel 3 times to create a (224 x 224 x 3) tensor, compatible with ImageNet-pretrained weights.
Step 5 — ImageNet normalisation	Mean = [0.485, 0.456, 0.406] Std = [0.229, 0.224, 0.225] (applied per channel to align with pretrained feature statistics).

Data Augmentation (training set only)

Augmentation	Parameter
Random horizontal flip	$p = 0.5$
Random vertical flip	$p = 0.5$
Random rotation	± 15 degrees

Paper sentence: "Raw CT slices were preprocessed using liver-specific HU windowing (-75 to 150 HU) followed by CLAHE (clip limit = 0.01) to enhance tumor boundaries. Images were resized to 224x224 and normalized using ImageNet statistics to enable transfer learning from the MobileNet-V2 backbone."

4. Dataset & Train/Val/Test Splits

Source: Kaggle — andrewmvd/liver-tumor-segmentation (Parts 1 & 2). 3D NIfTI volumes (.nii). 2D axial slices extracted from the middle 80% of each volume (top/bottom 10% discarded as anatomically empty).

Slice-level label assignment

Class	Rule
Class 0 — Background	Segmentation mask contains no liver or tumor (all zeros)
Class 1 — Liver	Segmentation mask contains label 1 (liver) but NOT label 2
Class 2 — HCC Tumor	Segmentation mask contains label 2 (tumor present)

Stratified split proportions

Split	Proportion
Train	72% (= 0.9 of data x 0.8 inner split)
Validation	18% (= 0.9 of data x 0.2 inner split)
Test	10% (held-out; stratified by class)

Note on class imbalance: Background slices typically outnumber Tumor slices significantly in liver CT datasets. Stratified splitting ensures each split preserves the original class proportions.

5. Loss Function

Loss function	CrossEntropyLoss (torch.nn.CrossEntropyLoss)
Why CrossEntropy	This is a 3-class classification task. CrossEntropy combines log-softmax and negative log-likelihood into one numerically stable computation — appropriate for mutually exclusive class labels.
Why NOT Dice loss	Dice loss is designed for pixel-wise segmentation masks. Since this model predicts a single label per slice (not per pixel), CrossEntropy is the correct choice.

Class imbalance handling	Stratified splits preserve class ratios. If needed, class weights can be passed to CrossEntropyLoss via the 'weight' parameter.
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Paper sentence: "CrossEntropyLoss was used to train the 3-class classifier, providing numerically stable optimization by combining log-softmax and negative log-likelihood in a single computation."

6. Evaluation Metrics

Metric	Formula / Description
Accuracy	(Correct predictions) / (Total samples) across all 3 classes
Dice Similarity Coefficient (DSC)	Per class: $2 X \cap Y / (X + Y)$. Reported for Background, Liver, Tumor, and Mean(Liver+Tumor).
Sensitivity (Recall)	$TP / (TP + FN)$. Primary clinical metric for tumor detection — minimises missed tumors.
Specificity	$TN / (TN + FP)$. Ability to correctly identify non-tumor slices.
Precision	$TP / (TP + FP)$. Fraction of predicted tumors that are true tumors.
F1 Score	$2 \times \text{Sensitivity} \times \text{Precision} / (\text{Sensitivity} + \text{Precision})$.

Clinical note: Sensitivity is the most critical metric in cancer detection — a false negative (missed tumor) is clinically more dangerous than a false positive. The training loop logs validation sensitivity every epoch.

7. Training Process & Callbacks

Training loop	Standard mini-batch SGD with backpropagation via Adam optimizer.
Gradient zeroing	optimizer.zero_grad() called at start of each batch.
Best model checkpoint	Saved to models/mobilenet_best.pth when validation accuracy improves.
LR scheduler	ReduceLROnPlateau: halves LR after 3 epochs of no val loss improvement.
Early stopping	NOT used — model trains for all 15 epochs.
Metrics logged/epoch	train_loss, val_loss, train_acc, val_acc, val_sensitivity, val_dice
Evaluation mode	model.eval() + torch.no_grad() used for validation and test phases.

Key training code (simplified)

```
criterion = nn.CrossEntropyLoss() optimizer = torch.optim.Adam(model.parameters(),
lr=0.001) scheduler = ReduceLROnPlateau(optimizer, mode='min', factor=0.5, patience=3) for
epoch in range(1, 16): model.train() for images, labels in train_loader:
```

```
optimizer.zero_grad() outputs = model(images) loss = criterion(outputs, labels)
loss.backward() optimizer.step() scheduler.step(val_loss) # reduce LR if plateau
```

Paper sentence: "The model was trained using backpropagation with the Adam optimizer (lr=0.001). A learning rate scheduler reduced the LR by 50% upon validation loss plateau, and the best model weights were retained based on peak validation accuracy."

8. Model Size & Performance Profile

Property	Value
Total parameters	~3.5 million (standard MobileNet-V2 + custom head)
Backbone parameters	~3.4 million (MobileNet-V2 inverted residual blocks)
Head parameters	~0.33 million (Linear 1280->256 + Linear 256->3)
All layers trainable	Yes — full end-to-end fine-tuning
Model file size	~14 MB (.pth weights file)
Input tensor size	1 x 3 x 224 x 224 (batch x channels x H x W)
Deployment suitability	Lightweight — suitable for edge/clinical hardware (mobile claim valid)

Paper sentence: "The lightweight MobileNet-V2 backbone contains approximately 3.5 million parameters, yielding a model file of ~14 MB — demonstrating suitability for deployment on resource-constrained clinical hardware."

9. Unique Contributions / Proposed Method

Contribution	Details
Domain-specific preprocessing	HU windowing [-75, 150] + CLAHE (clip=0.01) pipeline specifically tuned for liver CT. This is NOT standard ImageNet preprocessing — it exploits radiological domain knowledge.
Anatomically-aware slice selection	Only the middle 80% of axial slices per volume are used. The top/bottom 10% are discarded as they contain no liver tissue, reducing noise in the training set.
Regularized two-layer classifier head	Two-stage Dropout (0.2 before first FC, 0.3 before output) + intermediate 256-d projection layer, instead of the standard single linear layer in MobileNet-V2.
Clinical sensitivity-first evaluation	Validation sensitivity is logged every epoch alongside accuracy, prioritizing tumor recall over overall accuracy. Includes full case analysis (TP, FN, FP breakdown).

Paper sentence: "The proposed method introduces a liver-specific preprocessing pipeline integrating HU windowing and CLAHE prior to feature extraction, combined with a regularized two-layer classification head to improve

generalization on the HCC dataset."

10. Challenges & Limitations

Challenge	Description
Class imbalance	Background slices vastly outnumber Tumor slices in CT volumes. Stratified splits preserve proportions but do not eliminate the imbalance effect on loss.
Weak supervision / slice-level labels	If a slice contains even one tumor pixel in the segmentation, the whole slice is labelled Tumor. The model never receives pixel-level supervision.
Loss of 3D context	2D slice classification discards the inter-slice spatial relationships available in the full 3D volume, potentially missing tumor boundaries at the edges of lesions.
Grayscale-to-RGB conversion	Stacking grayscale channels to form a fake RGB image is a workaround for ImageNet-pretrained weights. The three channels carry identical information.
Training on limited cases	Code shows max_cases=5 (5 NIfTI volumes) for initial runs. Results depend heavily on the total number of volumes actually processed.

Paper sentence: "The model initially suffered from class imbalance between background and tumor slices, and the weak slice-level labeling strategy may introduce label noise for slices containing only marginal tumor regions."

11. Graph Data Available from Training

The training loop populates a **history** dictionary with the following tracked values per epoch. These are the recommended graphs to include in your paper:

Plot	Importance
train_loss / val_loss vs epoch	PRIMARY GRAPH — shows convergence. Divergence between train/val loss reveals overfitting.
train_acc / val_acc vs epoch	PRIMARY GRAPH — shows accuracy progression over training.
val_sensitivity vs epoch	Shows how tumor recall evolves — clinically critical metric.
val_dice vs epoch	Mean DSC for Liver+Tumor classes per epoch.
Confusion matrix	Saved to output/confusion_matrix.png — normalized and raw count versions.
Clinical dashboard	Saved to output/clinical_dashboard.png — pie chart, heatmap, probability distributions.

Tumor case analysis	Saved to output/tumor_case_analysis.png — TP, FN, FP visual breakdown.
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12. Master Checklist — All Items Resolved

Checklist Item	Value
Input size	224 x 224 x 3
Architecture	MobileNet-V2 backbone + custom 2-layer FC head
Task type	Slice-level CLASSIFICATION (not pixel segmentation)
Loss function	CrossEntropyLoss
Optimizer	Adam
Learning rate	0.001
Batch size	32
Epochs	15
LR scheduler	ReduceLROnPlateau (factor=0.5, patience=3)
Dataset split	72% train / 18% val / 10% test (stratified)
Augmentation	H-flip (p=0.5), V-flip (p=0.5), Rotation +/-15 deg (train only)
Preprocessing	HU windowing [-75,150] + CLAHE (clip=0.01) + ImageNet norm
Metrics	Accuracy, DSC, Sensitivity, Specificity, F1, Confusion Matrix
Model parameters	~3.5 million
Model file size	~14 MB
Best model saved	models/mobilenet_best.pth (best val accuracy)
Graphs available	Loss, Accuracy, Sensitivity, DSC, Confusion Matrix, Dashboard
Unique contribution	Domain-specific preprocessing + regularized head + sensitivity-first eval